

Lab 6: Control using Git and GitHub

Objectives:

- To understand the concept of version control systems.
- To learn basic Git commands and operations.
- To create and manage repositories using GitHub.
- To perform push, pull, clone, and commit operations.
- To understand collaboration and source code management.
- To track project changes using Git and GitHub.

Theory:

Git is an open-source distributed version control system used to track changes in source code during software development. It helps developers manage project versions, collaborate with team members, and maintain project history efficiently.

GitHub is a cloud-based platform that hosts Git repositories and provides collaboration features such as code sharing, issue tracking, pull requests, and project management.

Git and GitHub are widely used in Software Project Management because they:

- Maintain source code history.
- Support team collaboration.
- Allow rollback to previous versions.
- Manage software development workflows.
- Enable remote repository hosting.

Basic Git Concepts:

a. Repository

A storage location that contains project files and version history.

b. Commit

A saved snapshot of project changes.

c. Push

Uploads local repository changes to GitHub.

d. Pull

Downloads latest changes from GitHub repository.

e. Clone

Copies a remote repository to local computer.

f. Branch

A separate development line used for new features or testing.

Advantages of Git and GitHub

- Easy version tracking
- Team collaboration support
- Remote backup of projects

- Fast and lightweight
- Open-source and widely used
- Supports Agile development workflow

Common Git Commands

Command	Description
git init	Initialize local repository
git clone	Copy repository from GitHub
git status	Check repository status
git add .	Add files for commit
git commit -m "message"	Save changes
git push	Upload changes to GitHub
git pull	Download latest changes
git log	View commit history

Steps / Procedure

1. Initialize Git Repository

Initializes a local Git repository in the project folder.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS
● PS F:\7th-Sem-23081003\SPM\Lab7Git> git init
  Initialized empty Git repository in F:/7th-Sem-23081003/SPM/Lab7Git/.git/
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

2. Check Git Status

Displays the current status of files and repository.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS
● PS F:\7th-Sem-23081003\SPM\Lab7Git> git status
  On branch main

  No commits yet

  nothing to commit (create/copy files and use "git add" to track)
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

3. Create README File

Creates a README.md file for the repository.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS
● PS F:\7th-Sem-23081003\SPM\Lab7Git> echo "# lab7GitSPM" >> README.md
● PS F:\7th-Sem-23081003\SPM\Lab7Git> ls

  Directory: F:\7th-Sem-23081003\SPM\Lab7Git

  Mode                LastWriteTime         Length Name
  ----                -
  -a----             5/9/2026   9:42 PM             30 README.md

○ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

4. Add Files to Staging Area

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS
● PS F:\7th-Sem-23081003\SPM\Lab7Git> git add .
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

5. Commit Changes

Saves the staged files into local repository with a commit message.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git commit -m "first commit"
[main (root-commit) c8ffc5d] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 README.md
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

6. Rename Branch to Main

Renames the current branch to main.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git commit -m "first commit"
[main (root-commit) c8ffc5d] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 README.md
● PS F:\7th-Sem-23081003\SPM\Lab7Git> git branch -M main
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

7. Connect Local Repository with GitHub

Connects local repository with remote GitHub repository.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS  powershell + v |

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git remote add origin https://github.com/sulav063/lab7GitSPM.git
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

8. Push Files to GitHub

Uploads local repository files to GitHub.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 237 bytes | 237.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   https://github.com/sulav063/lab7GitSPM.git
To https://github.com/sulav063/lab7GitSPM.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

9. Pull Latest Changes

Downloads latest updates from GitHub repository.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git pull origin main
From https://github.com/sulav063/lab7GitSPM
 * branch          main          -> FETCH_HEAD
Already up to date.
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> 
```

10. View Commit History

Displays commit history of repository.

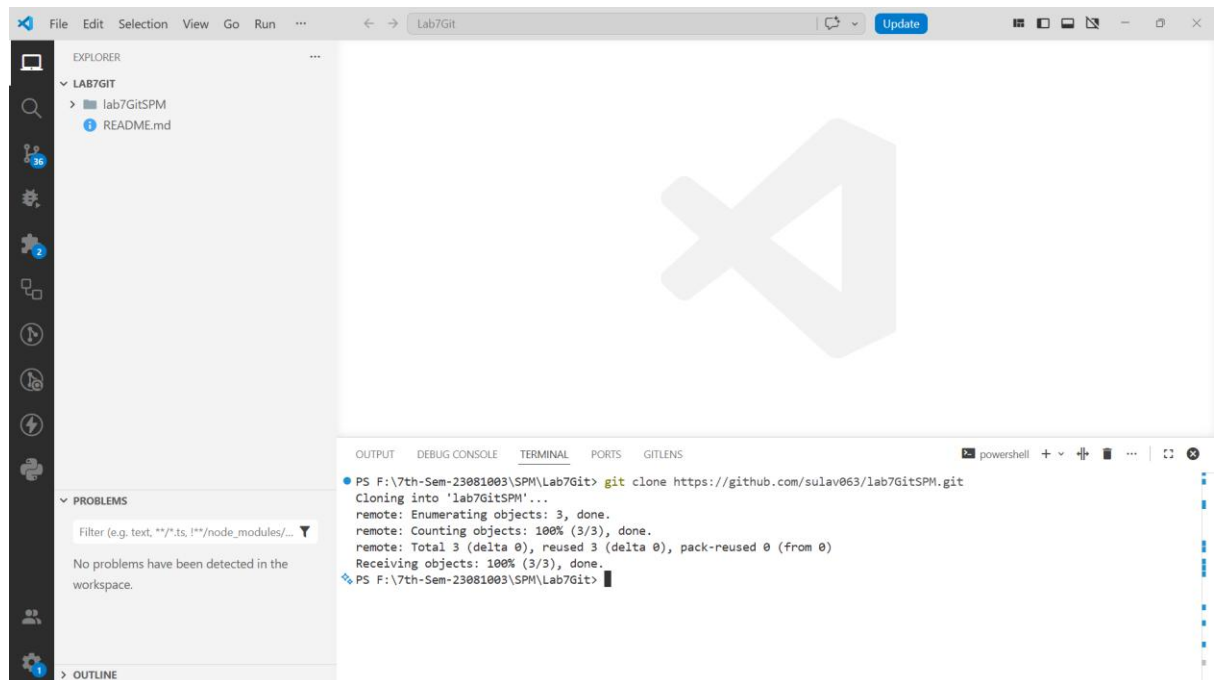
```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git log
commit c8ffc5d1d5094b66e80f4e284dfe1d6fb2c69a4d (HEAD -> main, origin/main)
Author: sulav063 <sulav057@gmail.com>
Date:   Sat May 9 21:45:19 2026 +0545

    first commit
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> 
```

11. Clone Repository

Creates a copy of GitHub repository in local computer.



12. Create a New Branch

Creates a new branch named feature-login.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git branch feature-login
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

13. View All Branches

Displays all available branches in the repository.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git branch
    feature-login
*   main
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

14. Switch to New Branch

Switches from main branch to feature-login branch.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git checkout feature-login
Switched to branch 'feature-login'
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git checkout feature-login
Switched to branch 'feature-login'
● PS F:\7th-Sem-23081003\SPM\Lab7Git> git branch
*   feature-login
    main
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

15. Create a New File in Branch

Creates a new file named login.txt inside the branch.

- PS F:\7th-Sem-23081003\SPM\Lab7Git> echo "Login Module" >> login.txt
- PS F:\7th-Sem-23081003\SPM\Lab7Git> ls

Directory: F:\7th-Sem-23081003\SPM\Lab7Git

Mode	LastWriteTime	Length	Name
d----	5/9/2026 9:50 PM		lab7GitSPM
-a----	5/9/2026 9:55 PM	30	login.txt
-a----	5/9/2026 9:42 PM	30	README.md

- PS F:\7th-Sem-23081003\SPM\Lab7Git> █

16. Check Repository Status

Shows modified and newly created files.

OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

- PS F:\7th-Sem-23081003\SPM\Lab7Git> git status

On branch feature-login
Untracked files:
 (use "git add <file>..." to include in what will be committed)

lab7GitSPM/
login.txt

nothing added to commit but untracked files present (use "git add" to track)

- ✚ PS F:\7th-Sem-23081003\SPM\Lab7Git> █

17. Add File to Staging Area

OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

- PS F:\7th-Sem-23081003\SPM\Lab7Git> git add .

warning: adding embedded git repository: lab7GitSPM
hint: You've added another git repository inside your current repository.
hint: Clones of the outer repository will not contain the contents of
hint: the embedded repository and will not know how to obtain it.
hint: If you meant to add a submodule, use:
hint:
hint: git submodule add <url> lab7GitSPM
hint:
hint: If you added this path by mistake, you can remove it from the
hint: index with:
hint:
hint: git rm --cached lab7GitSPM
hint:
hint: See "git help submodule" for more information.
hint: Disable this message with "git config set advice.addEmbeddedRepo false"- ✚ PS F:\7th-Sem-23081003\SPM\Lab7Git> █


```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git commit -m "Added login module file"
[feature-login 36f3de3] Added login module file
 2 files changed, 1 insertion(+)
 create mode 160000 lab7GitSPM
 create mode 100644 login.txt
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

18. Switch Back to Main Branch

Returns to the main branch.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git checkout main
warning: unable to rmdir 'lab7GitSPM': Directory not empty
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

19. Merge Branch with Main

Merges changes from feature-login branch into main branch.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git merge feature-login
Updating c8ffc5d..36f3de3
Fast-forward
 lab7GitSPM | 1 +
 login.txt  | Bin 0 -> 30 bytes
 2 files changed, 1 insertion(+)
 create mode 160000 lab7GitSPM
 create mode 100644 login.txt
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

20. Push Updated Repository

Uploads updated repository to GitHub.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 348 bytes | 348.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:  https://github.com/sulav063/lab7GitSPM.git
To https://github.com/sulav063/lab7GitSPM.git
    c8ffc5d..36f3de3  main -> main
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

21. Delete Branch

Deletes the branch after merging.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git branch -d feature-login
Deleted branch feature-login (was 36f3de3).
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

22. View Remote Repository

Displays the connected remote GitHub repository URLs.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

● PS F:\7th-Sem-23081003\SPM\Lab7Git> git remote -v
origin  https://github.com/sulav063/lab7GitSPM.git (fetch)
origin  https://github.com/sulav063/lab7GitSPM.git (push)
❖ PS F:\7th-Sem-23081003\SPM\Lab7Git> █
```

Conclusion:

In this lab, Git and GitHub were used for version control and repository management. Basic Git operations such as repository creation, commit, push, pull, and clone were performed successfully. The lab provided practical knowledge of collaborative software development and source code management using Git and GitHub.